

Data Mining: Report

Predicting the selectivity of a Parcoursup programme

Parcoursup 2020 dataset

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1. Problem definition and objectives

Research question: can we predict the degree of selectivity of a Parcoursup programme measured by its *access rate* from its structural characteristics and the profile of its applicants?

The access rate (ratio between the last called and the last ranked applicant) is chosen as the target rather than the raw last-called rank, because it is normalised across programmes of very different sizes and is therefore directly comparable. Two objectives are pursued:

- Primary objective (regression): estimate the access rate of each programme from variables known before the admission outcome, which rules out any risk of data leakage. Several regression models are built and compared to select the most accurate one.
- Secondary objective (clustering): group programmes by admission profile (rank, capacity, scholarship rate, honours distribution) to reveal interpretable typologies and qualitatively inform the regression.

2. Data and exploratory analysis

The raw Parcoursup 2020 dataset contains 114 variables and 12,760 programmes, for a single year. Exploratory analysis covered distributions, correlations and missing values, and guided the selection of variables retained for each of the two tasks.

The extreme values identified (CNED capacities, high ranks of schools with common competitive exams, CPGE with 100% “Very Good” honours) correspond to real, contextually coherent cases. They were therefore not removed but handled through log transformation at the normalisation stage.

3. Data preparation

3.1 Cleaning

365 rows (2.86%) were removed, split across three verified causes: programmes with no admits (`acc_tot` = 0, 65 rows), programmes with no admitted new graduates (`acc_neobac` = 0, 94 rows) and programmes under a grouped common competitive exam with no individual rank (`ran_grp1` missing, 229 rows, mainly “Other programme”). No exact duplicates were found. The residual bias, slight under-representation of “Other programmes” (−1.66 pts), is documented.

3.2 Feature engineering

Several variables were engineered:

- pressure = applicants / capacity: how many applicants compete for each seat, the expected strongest predictor.
- prct_filles, prct_bac_gen, prct_boursiers: shares of female applicants, general-track baccalaureate holders and scholarship holders, describing the demand profile.
- fill rate (admits / capacity), replacing the admit count to break its collinearity with capacity (correlation reduced from 0.94 to 0.05–0.15).
- zone: regions grouped into Île-de-France / Rest of France / Overseas-Abroad, with overseas isolated because its selectivity context is structurally different.

3.3 Encoding and normalisation

The skewed variables capacite and ran_grp1 were log-transformed (skewness reduced from 11.10 and 6.46 to 0.73 and 0.30). One-hot encoding (drop_first) was applied to the categorical variables (status, zone, field), selectivity was binary-encoded, and a StandardScaler was applied to the continuous variables. The final clustering matrix has 12,395 rows $\times$ 22 columns, with no missing values and no variable pair correlated above 0.8. Identifiers (institution, field, zone, status, region) are kept separate to allow the clusters to be read and named.

4. Modelling: Regression

4.1 Protocol and evaluation

Since the target (taux_acces) is numerical and continuous, three regression models were built and compared: Linear Regression, k-Nearest Neighbours (kNN) and Random Forest. The goal of the comparison was to identify the method producing the most accurate predictions while retaining the interpretability provided by linear regression.

All three models were trained in Orange on 12,526 observations and 19 explanatory variables (Establishment as a meta variable; the text variable Selectivity ignored because it duplicates Selectivite_bin), and evaluated under the same 10-fold cross-validation so their performances can be compared fairly.

4.2 Model comparison

Model	R ²	MAE	RMSE	MAPE	sMAPE
Linear Regression	0.601	13.670	17.579	46.16%	32.92%
kNN	0.638	12.243	16.749	38.97%	27.82%
Random Forest	0.783	9.312	12.966	26.92%	21.92%

For MAE, RMSE, MAPE and sMAPE, lower values indicate better performance; for R², higher is better. Random Forest ranks first on every metric, followed by kNN, with Linear Regression last. Compared with Linear Regression, Random Forest reduces the MAE by $\approx 32\%$ (13.67 $\rightarrow$ 9.31), reduces the RMSE by $\approx 26\%$ (17.58 $\rightarrow$ 12.97) and raises the R² from 0.601 to 0.783, the model explains an additional 18 points of variance. These gaps suggest that the relationship between the explanatory variables and the access rate is not entirely linear: Random Forest captures non-linear

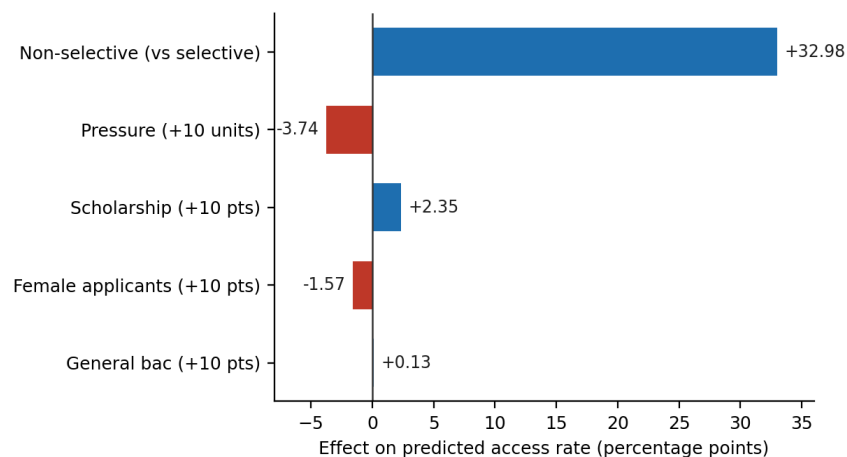
effects and interactions (e.g. how pressure interacts with programme category or region) that a global linear model cannot represent.

4.3 Final model selection

Random Forest is retained as the final predictive model, because it achieves the best score on every metric. kNN improves modestly on Linear Regression by predicting from similar programmes rather than a global relationship, but the gain remains limited compared with Random Forest. Linear Regression is nevertheless kept as an interpretable baseline, because its coefficients allow the direction and magnitude of each variable's association with the access rate to be read directly, something Random Forest cannot produce as a compact summary. The two approaches therefore play complementary roles: Linear Regression supports interpretation, Random Forest delivers accuracy.

4.4 Interpretation of the Linear Regression baseline

The Linear Regression coefficients describe statistical associations, not causal effects. A positive coefficient means the variable is associated with an increase in the predicted access rate; a negative coefficient with a decrease, all else being equal.



Effect of each variable on the predicted access rate (all else being equal).

- Selectivity (dominant effect): coefficients are +16.489 (non-selective) and -16.489 (selective), a gap of ≈ 33 points. All else equal, a non-selective programme has a predicted access rate about 33 points higher than a selective one. This is consistent with more restrictive admission conditions.
- Pressure: coefficient -0.374; +1 unit of pressure $\rightarrow$ -0.37 point of access rate (i.e. -3.74 for +10 units). Programmes in high demand relative to their capacity are harder to enter, the expected effect is confirmed.
- Scholarship share: coefficient +23.526; +10 points of scholarship applicants $\rightarrow$ +2.35 points. A statistical association, not causal, likely partly reflecting the fact that non-selective programmes tend to have both higher access rates and a broader social recruitment.
- Female-applicant share: coefficient -15.697; +10 points $\rightarrow$ -1.57 point of access rate.
- General-baccalaureate share: coefficient +1.254; a small effect (+0.13 for +10 points) compared with selectivity, pressure and the scholarship share.

The categorical variables (region, status, field), turned into indicators, show coefficients of varied signs: access rates vary with location and programme type, but these effects should be read cautiously since some categories rest on few observations. Random Forest's superior performance suggests that these effects also interact with each other.

5. Modelling: Clustering

Two unsupervised approaches were compared in Orange on the normalised matrix, aiming to surface programme typologies based on their admission profile (rank, volume, social and academic profile of admits).

5.1 Protocol and comparative evaluation

- k-Means: normalised columns, 10 re-initialisations, 300 iterations; the number of clusters was optimised from 5 to 7 on the Silhouette Plot. Scores are 0.146 (k=5), 0.151 (k=6) and 0.149 (k=7): k = 6 maximises the silhouette and is retained.
- Hierarchical clustering: Ward linkage, dendrogram cut at Top N = 3 ($\approx$ 75% height ratio), validated in parallel by a Silhouette Plot.

Criterion	k-Means	Hierarchical (Ward)
Number of clusters	6	3
Silhouette	0.151	0.162 (weighted mean)

k-Means with 6 clusters is retained as the final solution. Hierarchical clustering serves as a comparison: the Silhouette Plot of the hierarchy hid a very poorly separated C3 cluster (per-cluster silhouettes: C1 = 0.266, C2 = 0.428, C3 = -0.003, see). Hierarchical clustering is kept as a coarser comparative view rather than as a competing solution.

5.2 Identified typologies (k-Means, 6 clusters)

Clusters were named from their average profiles (Group By on the normalised variables, mode on the categorical ones). A positive value indicates a level above the overall average, a negative one below. The split is dominated by short programmes (BTS), which explains the moderate silhouette: clusters differ mainly by profile (gender, scholarship, fill rate, honours) rather than by field.

Cluster	Distinctive profile	Label
C1	BTS, very few female applicants, small capacity	Technical BTS (male-dominated fields)
C2	Large capacity, high demand, Licence, non-selective	High-capacity university Bachelor
C3	BTS, high share of scholarship holders and women	Service-sector BTS, social profile
C4	Selective, CPGE, very high "Very Good" honours rate	Elite CPGE
C5	Very low fill rate (vacant seats)	Under-filled BTS
C6	Selective, high honours, few scholarship holders	Selective high-achieving BTS

5.3 Hierarchical typologies (Ward, 3 clusters)

The hierarchical solution, kept for comparison, was labelled using the same Group By on the average profiles of its three clusters. The Distances widget was set to Euclidean (consistent with Ward's variance-minimisation definition). The resulting split is approximately 45% / 10% / 45% (C1 / C2 / C3): unbalanced but interpretable, with C2 as a smaller, specialised cluster. The coarser split consolidates the six k-Means groups into higher-level typologies of the Parcoursup landscape.

Cluster	Distinctive profile	Label
C1 (~45%)	BTS, non-selective, small capacity, high scholarship share (+0.26), few high honours	Community BTS, social profile
C2 (~10%)	Large capacity (+0.98), high demand, Bachelor, non-selective, average honours	Mass university Bachelor
C3 (~45%)	Selective, CPGE mode, very high honours (pct_tb +0.62), few scholarship holders (-0.48)	Selective high-achieving programmes

Note on labelling: “CPGE” appears as the mode (most frequent field) of C3, not because every programme in this cluster is a CPGE. C3 more broadly groups selective, high-achieving programmes (CPGE, schools, IFSI...). The three groups roughly consolidate the six k-Means typologies along the axes short vs long, selective vs not, and honours-driven vs mass recruitment. Both views are complementary: the finer k-Means split captures nuances within short programmes (technical vs service-sector BTS, under-filled BTS), while the hierarchical split gives a clean high-level segmentation.

6. Interpretation of results

Several working hypotheses were formulated at the start of the project, some directly from the problem statement, others as reasonable a priori expectations about the drivers of the access rate. This section revisits each in light of the results.

H1: Pressure will likely be the strongest predictor. Partially rejected. Pressure has the expected direction and is significant (−0.37 point of access rate per +1 unit of pressure in the linear regression, and it is used by Random Forest), but it is not the dominant driver. **Declared selectivity has a much larger effect** (≈ 33-point gap between non-selective and selective programmes), making it the actual leading predictor. Pressure operates on top of that more fundamental structural distinction.

H2: CPGE and grande-école-type programmes concentrate the lowest access rates. Confirmed. Both clustering approaches isolate a cluster of selective, high-achieving programmes with a CPGE mode (C4 in k-Means, C3 in hierarchical), characterised by very high shares of “Very Good” honours the mark of hard-to-enter programmes. In the regression, however, the CPGE effect operates mainly *through the selectivity variable* rather than through the field indicator itself.

H3: Declared selectivity is consistent with observed access rates. Strongly confirmed. Non-selective programmes have a predicted access rate ≈ 33 points higher than selective ones, all else equal the largest effect in the entire regression. Both clustering approaches also cleanly separate the population along the selective / non-selective axis.

H4: The applicant profile (general-track bacs, scholarship holders) shapes the access rate. Confirmed but modest. Scholarship share carries the strongest profile effect (+2.35 points per +10 points of scholarship applicants) as a positive association likely because non-selective programmes tend to have broader social recruitment. General-track baccalaureate has almost no direct effect (+0.13 points), and the share of female applicants shows a small negative association (−1.57 points). These profile effects are real but an order of magnitude smaller than selectivity or pressure.

H5: Île-de-France vs the rest of France will have a notable effect. Weakly confirmed. Region coefficients are of varied signs and small magnitude relative to selectivity and pressure. Some regional variation exists, but geographic location is *not* among the primary drivers of the access rate at this level of aggregation.

H6: Expected clustering typologies: ultra-selective CPGE, high-capacity low-selectivity BTS, and regional IUT with strong local recruitment. Partially confirmed. The elite-CPGE typology and the high-capacity, low-selectivity typology both emerge (Bachelor rather than BTS for the latter, but the pattern matches). The “regional IUT” typology does not emerge: the clustering is dominated by BTS, and the finer k-Means split reveals within-BTS distinctions (technical vs service-sector, under-filled programmes) that were not part of the initial vision.

Overall. The results support the general logic of the initial framework, selectivity, pressure and applicant profile all drive the access rate in the expected directions, but reorder the importance of the drivers: selectivity dominates, pressure is a strong secondary effect, and profile characteristics are meaningful but smaller. The clustering surfaces two of the three expected typologies but reveals that within-BTS heterogeneity is the more informative segmentation axis for 2020, which the initial framework had not anticipated.

7. Limitations

- Single year 2020 (Covid context): no temporal dimension and no measure of robustness over time.
- Excluded programmes outside the standard ranking ($\approx 2.86\%$), leading to a slight under-representation of “Other programmes”.
- Unstable percentages on very small counts (100% honours driven by ≤ 5 admits), to be interpreted with caution.
- Linear Regression: $R^2 = 0.601$ (40% of variance unexplained); the target is bounded 0–100 but the linear model does not enforce it; linearity assumption and residual multicollinearity among volume variables that may make individual coefficients unstable.
- Random Forest: although it delivers the best accuracy ($R^2 = 0.783$), it is much harder to interpret than Linear Regression, its decision process cannot be summarised by a compact set of coefficients. Even so, $\approx 22\%$ of the variance still remains unexplained, pointing to relevant factors missing from the dataset (programme reputation, applicants’ grades, academic requirements, local demand, institution-specific admission policies, year-to-year changes in the number of seats).
- Clustering: moderate silhouette (0.151) and clusters dominated by BTS; some typologies rest on small counts.

8. Recommendations

- Integrate several years to improve robustness and add a temporal dimension.
- Enrich the data with variables likely to explain the remaining $\approx 22\%$ of variance: programme reputation, academic requirements, local demand, socio-economic indicators of the academy.
- Use SHAP values or permutation importance to open the Random Forest black box and quantify each variable's contribution to its predictions bridging the accuracy of Random Forest with the interpretability that only Linear Regression currently provides.
- Consider separate models per broad field if effects prove heterogeneous, and cross regression with clusters to target the segments where prediction error is highest.

9. Conclusion

The access rate of a Parcoursup programme can be predicted with good accuracy by a Random Forest model ($R^2 = 0.783$; $MAE \approx 9.3$ points), which outperforms kNN ($R^2 = 0.638$) and Linear Regression ($R^2 = 0.601$). The performance gap over Linear Regression confirms that the underlying relationships are partly non-linear and involve interactions between variables. Interpretation nevertheless relies on the Linear Regression baseline: the main driver is declared selectivity (≈ 33 -point gap between non-selective and selective programmes), followed by supply/demand pressure. Selectivity stems first from the ratio between applicants and capacity. The k-Means clustering complements this picture with six interpretable typologies of programmes, differentiated mainly by their social and academic profile. Random Forest is retained for prediction, Linear Regression for interpretation.